In [12]:
```
import numpy as np
a=[1,2,3,4,5,6]
b=np.copy(a)
print(b)
c=np.append(a,4)#add number at last
print(c)
d=np.insert(a,1,45)
print(d)
e=np.delete(a,2)
print(e)
# f=np.split(a,2)
# print(f)
a=[1,2]
b=[3,4]
np.array(a)
np.array(b)
np.concatenate((a,b))
```

```
[1 2 3 4 5 6]
[1 2 3 4 5 6 4]
[ 1 45  2  3  4  5  6]
[1 2 4 5 6]
```

Out[12]:
```
array([1, 2, 3, 4])
```

practice

In [13]:
```
a=[1,2,3,4,5]
arr=np.array(a)
print(arr)
```

```
[1 2 3 4 5]
```

In [16]:
```
#2
np.arange(1,21,1)
```

Out[16]:
```
array([ 1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11, 12, 13, 14, 15, 16, 17,
       18, 19, 20])
```

In [30]:
```
#3
import numpy as np
a=[1,2,3,4,5,6]
arr=np.array(a)
b=arr.reshape(2,3)
print(b)
# a = np.array([1, 2, 3, 4, 5, 6])
# b = a.reshape(2, 3)

# print(b)
```

```
[[1 2 3]
 [4 5 6]]
```

In [29]:
```
#4
a=[1,2,3,4]
arr=np.array(a)
print(arr)
b=arr.max()
print(b)
c=arr.argmax()
print(c)
```

```
[1 2 3 4]
4
3
```

saving and loadinig of numpy array

In [2]:
```
import numpy as np

arr=np.array([1,2,3,4])
np.save("data.npy",arr)
np.load("data.npy")
```

Out[2]:
```
array([1, 2, 3, 4])
```

In [5]:
```
a=np.array([1,2,3,4])
b=np.array([5,6,7,8])
np.savez("multiple.npz",a=a,b=b)
data=np.load("multiple.npz")
print(data["a"])
print(data["b"] )
```

```
[1 2 3 4]
[5 6 7 8]
```

In [17]:
```
a=[[1,2,3,4],[5,6,7,8],[1,3,7,9]]
arr=np.array(a)
print(arr)
print(arr[1,2])# 1 row and  or 2 column
print(arr[1])
print(arr[:,1])# row sari remove kerdeni hai or 1 column wali line dy ga
print(arr[1:3,1:3])
```

```
[[1 2 3 4]
 [5 6 7 8]
 [1 3 7 9]]
7
[5 6 7 8]
[2 6 3]
[[6 7]
 [3 7]]
```

In [18]:
```
arr=np.array([10,20,30,40])
arr>20
```

Out[18]:
```
array([False, False,  True,  True])
```

In [19]:
```
arr[arr>20]#filtering
```

Out[19]:
```
array([30, 40])
```

In [20]:
```
#broadcasting
arr=np.array([1,2,3])
arr+5
```

Out[20]:
```
array([6, 7, 8])
```

In [21]:
```
#2d broadcasting
arr=np.array([[1,2,3],
             [4,5,6]])
arr+np.array([10,20,30])
```

Out[21]:
```
array([[11, 22, 33],
       [14, 25, 36]])
```

In [22]:
```
a=input("enter your number")
arr=np.array(a)
print(arr)
```

```
[1,2,3,4]
```

In [ ]: